

Gauging Optimal Inventory For Ice Cream Flavors From Sales Data

Premise

An ice cream truck vendor can sell vanilla, chocolate, and strawberry ice cream to customers. They want to gauge the mean population preferences for those flavors in order to optimize both the inventory distribution and unit prices of the different flavors. Their plan to estimate and test the optimal ratio of vanilla, chocolate and strawberry ice cream to maintain in stock is to conduct a series of experiments:

1. Steady state (gauge overall favorite flavor of customers)
 - a. Assume there is no difference in preference, and maintain an equal ratio of all flavors
 - b. Keep prices constant
 - c. The flavor with the highest average units sold per day is predicted to be the population favorite, and vice versa
2. Gauge demand for each flavor and substitutability between the flavors (optimize prices)
 - a. Repeat the experiment in step 1 with various changes in the prices of each flavor, find out which price maximizes the revenue of each flavor (number of units sold times unit price)
 - b. Compute elasticities of demand for each flavor relative to price changes in other flavors
3. Test the optimal suite of demand as the ratio of flavor stock maintained through the simulation
 - a. Each optimal price in part 2 had a corresponding number of units sold
 - b. Use the ratio of those sales to estimate the optimal ratio to keep in stock
 - c. Compare the average daily profit against the steady state and other configurations tested in step 2
 - d. Report the results (success if the guess in (b) yielded the maximum profit)

Simulation Parameters

- Capacity of the ice cream truck: 300 units

- Inventory state of the truck (# of units of each flavor in stock), 100 units each initially, will be changed
- Unit cost of each flavor to the vendor: \$2.00/unit
- Unit price of each flavor to the customer: \$3.00/unit initially, will be changed
- Number of customers per day: normally distributed between 80 and 100
- Holding cost of unsold ice cream: \$1.00/unit per night

Person Parameters

- Flavor preferences (for vanilla, chocolate, and strawberry)
 - A probability distribution that is unknown (each must be at least 0.25, they sum to 1)
- Preference multipliers
 - Preferred flavor (highest value in distribution) is assigned 1, others are assigned $[\text{favorite_flavor_pref}]/[\text{other_flavor_pref}]$
 - This multiplier is applied to the unit price (cost of inconvenience of buying a different flavor than their favorite)
- Budget per order: \$10.00
- Diminishing returns coefficient (ϵ): product of values in probability distribution
 - Incentivizes people to diversify orders a little bit
- Ordering algorithm
 - Multiply flavor unit prices by preference multipliers for their personal analysis
 - Add the cheapest unit that is left in stock, subtract price from the budget, then multiply that price by ϵ and remove 1 unit of that flavor's stock
 - Repeat until no units are affordable (can never go over budget)

Simulation Flow (1 Trial)

- Given constant stock ratio to maintain and prices to customers
- Draw a number from the distribution of daily arrivals, execute that many orders, updating the stock of the ice cream truck
- Track total profit and units sold, both in general and for each flavor
- At the end of the day, subtract holding cost of all unsold units from profit
- Order more ice cream to refill stock according to the fixed stock ratio (subtract order cost from profit)
- Repeat for each subsequent day (simulate 30 days total)

Experiment Flow

1. 100 trials of the initial state
 - a. \$3.00 unit price of each flavor, equal ratio maintained in stock
 - b. Report 90% confidence interval for average daily sales of each flavor (first guess at which is the most preferred)
2. Price optimization
 - a. Vary the price of each flavor by different amounts (25% reduction to 25% increase), keeping the other flavor prices constant at \$3.00
 - b. For each trial, compute substitutability between the flavor whose price changed and each of the other flavors (i.e. $\% \text{change in price} / \% \text{change in sales of other flavor}$, relative to the mean values in the initial state)
 - c. Substitutability gives us more supporting evidence for our prediction for the population overall favorite flavor (low rates of substitution away from that flavor, high rates of substitution towards that flavor)
 - d. Conduct 100 trials, report 90% confidence intervals for those substitutability values
 - e. Save average daily sales for each flavor under each configuration over the 100 trials (no confidence interval for this one)
 - f. Use that data to estimate demand curves for each flavor, labeling the points (units sold, price) that resulted in the greatest individual profit for each flavor
3. Test optimal stock ratio
 - a. Compute optimal stock ratio from the optimal units sold for each flavor found in the previous step
 - b. Set prices to the optimal prices found in the previous step
 - c. Run 100 trials in that state, reporting a 90% confidence interval for the average daily profit (it should be higher than the statistic for the initial state, and higher than that for the other configurations in the price optimization phase)
 - d. If it is the greatest profit we've observed, success, otherwise, end the experiment anyway, reporting that we need to reconsider our guess or look for errors in the procedure